

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and ten years of experience building simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led an NSF-funded upgrade of the PatchworkMHD code and ran it on Frontera. Experienced in gradient-based calibration of physical models against observations. Currently building neural-network emulators of simulation output.

Skills

Languages: C/C++, Python, Fortran 95, Rust, R, Java, SQL, Julia, Shell

HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5, CPU and exploratory GPU parallel computing

Simulation: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Experience

Independent Contractor

Jan 2026 — Present

Mercor

- Provide deep expertise in physics and numerical methods to improve LLM models for a top AI lab.

Spatial Data Scientist

May 2025 — Jan 2026

TealWaters

- Led optimization of scientific code for topographic terrain modeling, improving computational performance of elevation derivative calculations.
- Engineered Python tools for geospatial data processing and analysis; applied random forest ML for wetland probability mapping.
- Performed EDA, feature engineering, and geospatial data preparation including multispectral (Sentinel-1/2) processing for ML/AI land-cover models.
- Collaborated with software engineers and scientific team to transition prototype simulation tools into production.

Software Engineer

Apr 2022 — Jan 2024

Paychex

- Developed Java Kafka consumers for high-throughput data transfer across enterprise databases; deployed to production via Jenkins and OpenShift using CI/CD pipelines.
- Conducted SQL data preparation, validation, and analysis from Oracle EBS datasets; built Splunk dashboards for production monitoring.
- Developed Python tests ensuring software quality and continuous integration standards.

Postdoctoral Research Associate

Feb 2021 — Apr 2022

Rochester Institute of Technology

- Led NSF-sponsored project to upgrade PatchworkMHD, a large-scale C-based HPC simulation code for modeling supermassive black hole binaries using state-of-the-art numerical techniques.
- Implemented new physics features (black hole spin) to the simulation library without impacting runtime, demonstrating expertise in large-scale scientific software architecture.
- Designed and ran large-scale computational experiments on the Frontera supercomputer (TACC), benchmarking and identifying performance optimization opportunities in parallel HPC code.
- Applied scientific machine learning (gradient-based optimization) to calibrate physical model parameters in Paramo against Fermi-LAT telescope observations, demonstrating ML-assisted simulation workflows for complex physical systems.
- Collaborated on multi-institutional HPC simulation project studying binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow

Oct 2018 – Nov 2020

Purdue University

- Architected and developed Paramo, an open-source multiscale/multiphysics simulation library in Fortran 95 modeling relativistic particle distributions via the Fokker-Planck equation coupled with radiative transfer across temporal and spatial scales; accelerated via OpenMP parallelization; explored GPU acceleration via OpenACC.
- Implemented advanced numerical methods including sophisticated PDE solvers, numerical integration, and particle distribution algorithms with arbitrary shape support, without impacting simulation runtime.
- Designed Python analysis and visualization tools interfacing with the Paramo simulation library, enabling statistical modeling and parameter optimization workflows.
- Obtained and led a NASA grant (\$68K) to model radiation from active galaxies; managed 3 PhD researchers and published 2 peer-reviewed papers.

Postdoctoral Research Fellow

Jan – Sep 2018

Universidad Michoacana de San Nicolás de Hidalgo

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant

Oct 2011 – Jul 2017

Universitat de València

- Independently developed Shell and Python pipelines to run the C-SPEV radiative transfer simulation code and process large HDF5 output datasets.
- Implemented numerical tools and data handling extensions to C-SPEV for computing both discrete and continuous spectra from particle distributions with arbitrary shape.
- Published 2 first-author papers on numerical simulations of relativistic plasma dynamics.

Projects

Blazar Population ML, github.com/altjerue/ilwikak-ml.

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares using a general-purpose time series foundation model, fine-tuned with Tleco-generated synthetic light curves.

WindsOfChange, github.com/UW-ForestMycobiomeLab/WindSporesHills.

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++)

- Developed C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, github.com/zkdavis/Tleco.

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, github.com/altjerue/paramo.

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONAHCYT Funded)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

Selected Publications (9 total, 2 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, *ApJ* **976**, 182, (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, *MNRAS* **501**, 4092–4102, (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, *MNRAS* **468**, 1169–1182, (2017), arXiv:1612.06383.

Education

Universitat de València

Valencia, Spain

Ph.D. in Physics. Excellent Cum Laude.

Jul. 2017

Universidad Michoacana de San Nicolás de Hidalgo

Morelia, Mexico

M.Sc. in Physics.

Universidad Autónoma del Estado de México

B.Sc. in Physics.

Sep. 2011

Toluca, Mexico

Dec. 2008